

ATPL EASA STUDY GUIDE | 032 - PERFORMANCE

Chapter 8: Class A En-Route Engine-Out Drift Down & Obstacles

Syllabus Source: CAE Oxford (pp. 283-316) | Pass Target: $\geq 90\%$ AviationExam

1. THE ENGINE FAILURE IN CRUISE PHYSICS

When an engine fails during cruise at high altitude (e.g. FL 370), the aircraft cannot maintain altitude because drag at that level exceeds maximum continuous thrust (MCT) of the remaining engine:

- **Drift Down Procedure:** The aircraft trades potential energy (altitude) for kinetic energy (airspeed), descending gradually at optimal single-engine speed (V_{md} / LRC speed) to maximize distance flown during descent toward the stabilizing ceiling.
- **Stabilizing Altitude (OEI Ceiling):** The level altitude where remaining engine thrust at MCT exactly balances aircraft drag. As fuel burns off and the aircraft gets lighter, the stabilizing ceiling gradually climbs!
- **Net En-Route Flight Path (1.1% Deduction):** Under CS-25 for twin-engine aircraft, the net en-route flight path is derived by deducting a 1.1% climb gradient penalty from gross data (1.4% for 3-engine, 1.6% for 4-engine).

2. EN-ROUTE OBSTACLE CLEARANCE REQUIREMENTS

Under EASA AIR-OPS (CAT.POL.A.215), the one-engine-inoperative net drift-down flight path must clear all high terrain and obstacles within 10 NM of the route centerline by AT LEAST 2,000 FT VERTICALLY!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): En-Route Driftdown Obstacle Clearance

Driftdown Procedure: Step 1: Set MCT (Maximum Continuous Thrust) on live engine. Step 2: Pitch to driftdown speed (approx L/D max) and disconnect altitude hold. Step 3: Turn 90 degrees off airway centerline per ATC oceanic/enroute contingency. Step 4: EASA Obstacle Margin: Net driftdown path must clear all terrain within 10 NM of track by at least 2,000 ft! Step 5: If single-engine ceiling cannot clear terrain, designate pre-computed escape routes to divert fields.

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Verifying Drift Down Terrain Clearance

SCENARIO (Alpine Mountain Crossing Drift Down Drill): A twin-jet airliner is cruising at FL 350 over the Alps: - An engine fails at point A - The OEI drift-down table shows that the net flight path passes through 14,500 ft MSL at the critical ridge - The highest mountain peak within 10 NM of track has an elevation of 12,200 ft

MSL QUESTION: Is the drift-down flight path legal under EASA CAT.POL.A.215 regulations? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Identify the mandatory EASA terrain clearance margin: - Required Vertical Clearance = 2,000 ft above terrain within 10 NM of track. Step 2: Calculate the Minimum Allowable Net Flight Path Altitude: - Min Net Altitude = Peak Elevation + Required Clearance - Min Net Altitude = 12,200 ft + 2,000 ft = 14,200 ft MSL. Step 3: Compare Actual Net Altitude with Minimum Required Altitude: - Aircraft Net Altitude at Ridge = 14,500 ft MSL - Minimum Required Altitude = 14,200 ft MSL - Actual Vertical Margin = 14,500 ft - 14,200 ft = 300 ft! Step 4: Check Compliance: - 300 ft > 2,000 ft legal requirement -> COMPLIANT! FINAL ANSWER: Yes, the flight path is legal. The aircraft clears the peak with 300 ft net margin, exceeding the 2,000 ft regulatory requirement.

[!] EXAM TRAP (AVIATIONEXAM) - The 10 NM Corridor & 2,000 ft Margin Trap

- Obstacle search corridor width is 10 NM ON EITHER SIDE of track (20 NM total width). - Clearance requirement is 2,000 ft (NOT 1,000 ft!). 1,000 ft is only permitted if the aircraft has reached its level stabilizing altitude and navigational accuracy is within 95%.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: Under EASA regulations, by what vertical margin must the OEI NET flight path clear all obstacles within 10 NM?

- [A]: 500 ft
- [B]: At least 2,000 ft vertically
- [C]: 1,000 ft
- [D]: 35 ft

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. CAT.POL.A.215 requires the one-engine-inoperative net flight path to have a positive slope at least 2,000 ft above all terrain within 10 NM on each side of the intended track.

Q2: What is the regulatory gradient penalty applied to gross drift down performance to obtain the NET drift down gradient?

- [A]: 0.5%
- [B]: 1.1% gradient deduction
- [C]: 2.0%
- [D]: Zero

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. CS-25 requires deducting 1.1% from gross climb/descent gradient data for twin-engine aircraft during the en-route phase.

Q3: What speed should the flight crew fly to maximize distance traveled during an engine-out drift down?

- [A]: Maximum operating Mach (Mmo)
- [B]: The single-engine drift down speed (approx Vmd / Long Range Cruise speed)
- [C]: Stall speed
- [D]: Minimum clean speed

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Flying at the speed for minimum drag (Vmd) yields the maximum lift-to-drag ratio (L/D max), resulting in the shallowest glide angle and maximum distance to the OEI ceiling.